AI Gym & Fitness Assistant

Documentation & Testing Report

Name: Kashish Sharma

1. Introduction

This project is about developing an AI-based Gym and Fitness Assistant that helps users manage their workouts, diet, and fitness habits in one place. Nowadays, people use different apps for exercise tracking, diet planning, and motivation. This project combines all these features into a single system using AI and machine learning.

2. Objective

The main objective of this project is to:

- Provide a smart fitness assistant

- Help users track workouts and diet

- Improve consistency and motivation

- Use AI to give personalized suggestions

3. Project Overview

The system is divided into different modules, where each module performs a specific task like workout tracking, diet planning, etc. The backend is developed using Python and FastAPI, and all features are connected through APIs.

4. Technologies Used

Frontend: Streamlit (Python)

Backend: Python, FastAPI

Machine Learning: Scikit-learn

Pose Detection: YOLOv8

Platform: Azure functions, Azure app service

Visualization: Matplotlib, Plotly

Deployment: Google Colab

5. System Architecture

The system follows a modular approach, where each module works independently but is connected through a common backend.

Flow:

1. User sends request

2. Request goes through API

3. Backend processes it

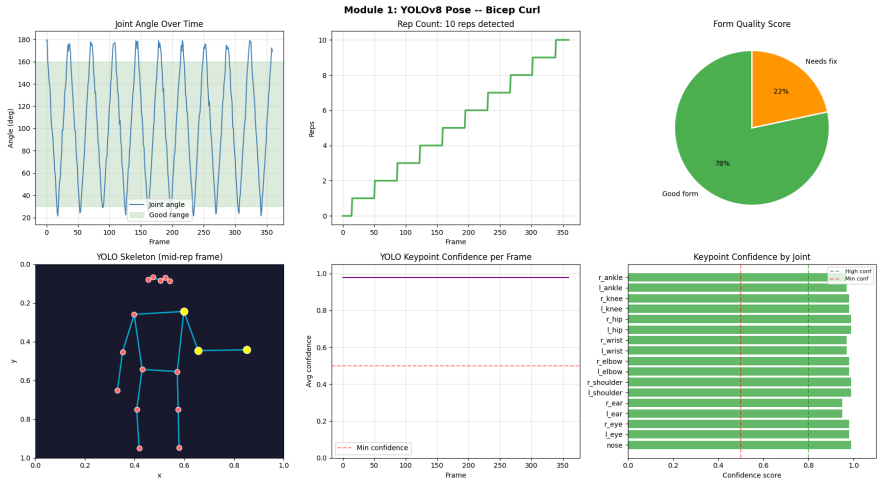
4. AI model generates result

5. Response is sent back to user

6. Modules Description

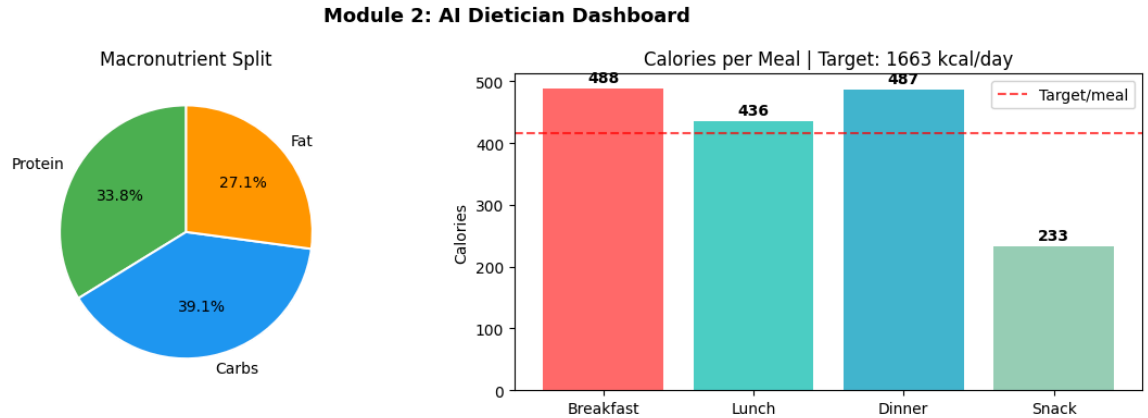
6.1 AI Gym Trainer

Detects exercises using pose detection, counts repetitions, and gives feedback on form.



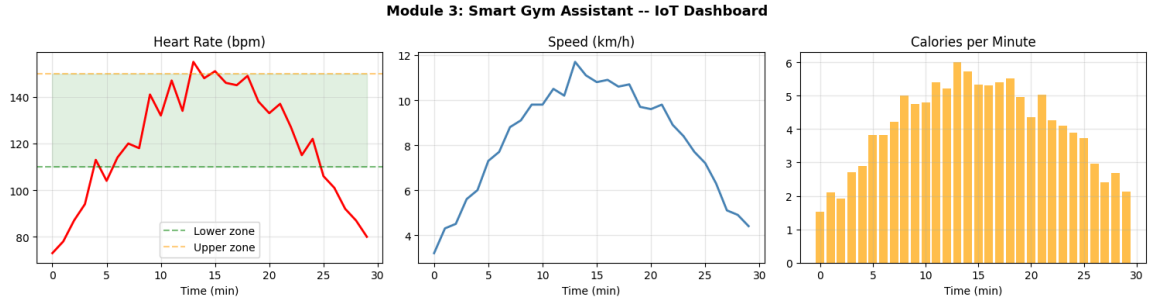
6.2 AI Dietician

Calculates BMI, BMR, TDEE and generates meal plans with calorie suggestions.



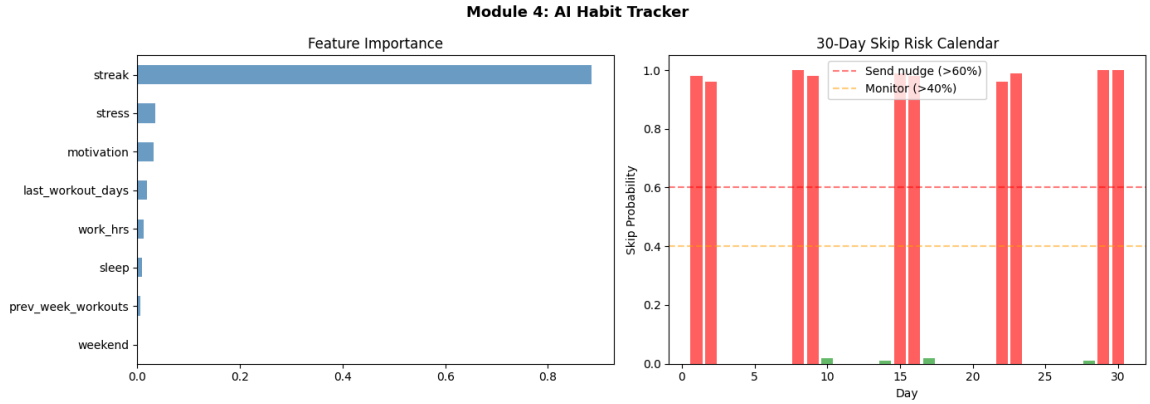
6.3 Smart Gym Assistant (IoT)

Simulates gym equipment data, tracks heart rate and calories, and provides workout suggestions.



6.4 Habit Tracker

Predicts if user may skip workout using machine learning and gives motivational suggestions.



6.5 Virtual Gym Buddy

Chat-based assistant that understands user messages and gives fitness and diet advice.

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Virtual Gym Buddy -- Demo Conversation
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User  : I feel so lazy today, I do not want to go to the gym
Intent: motivation | Sentiment: negative
Buddy : I hear you! You have already done the hardest part -- deciding to start!

User  : My legs are sore and tired from yesterday
Intent: tired | Sentiment: negative
Buddy : I hear you! A lighter workout today is still better than skipping. Try 20 minutes?

User  : What should I eat after my workout?
Intent: diet | Sentiment: neutral
Buddy : Protein within 30 min post-workout really helps recovery.

User  : I am not seeing any progress, it is really frustrating
Intent: progress | Sentiment: neutral
Buddy : Plateaus are normal! Try changing your routine every 4-6 weeks.

User  : I love my workout today, feeling amazing!
Intent: default | Sentiment: positive
Buddy : I am here to support your fitness journey! Ask me about workouts, diet, or motivation!

User  : My knee hurts during squats
Intent: injury | Sentiment: negative
Buddy : I hear you! Modify your workout around the injury.

Daily Quote: Your only competition is who you were yesterday.
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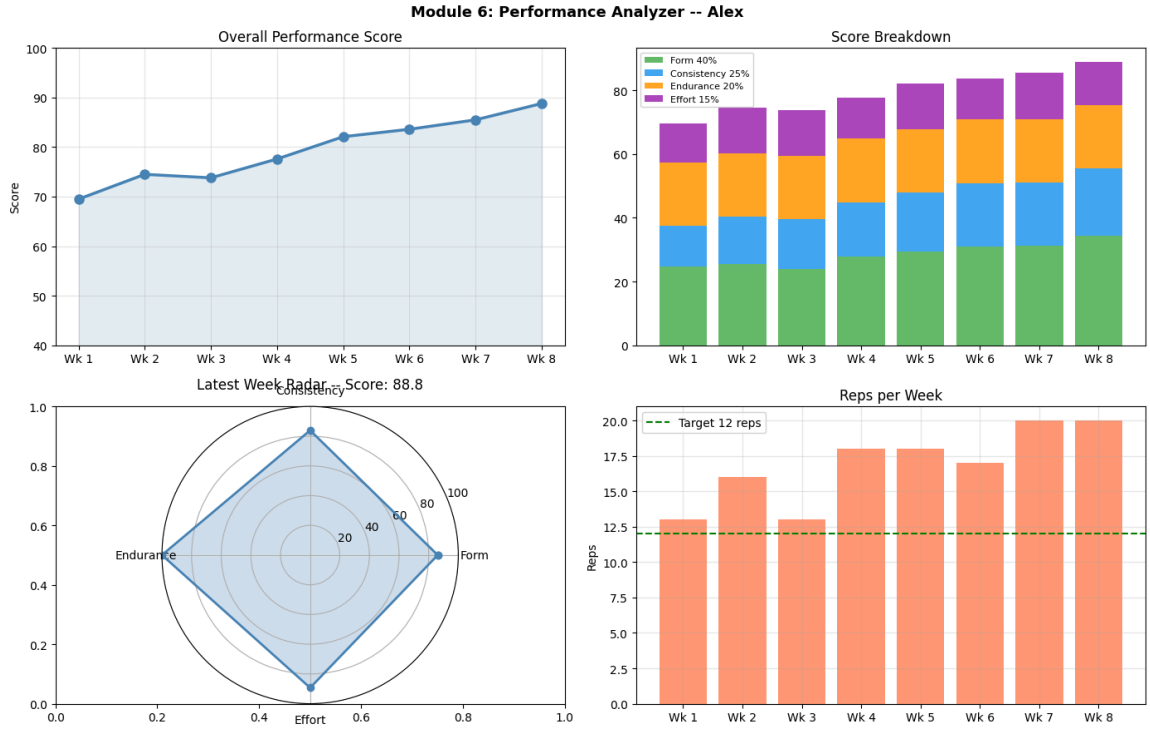
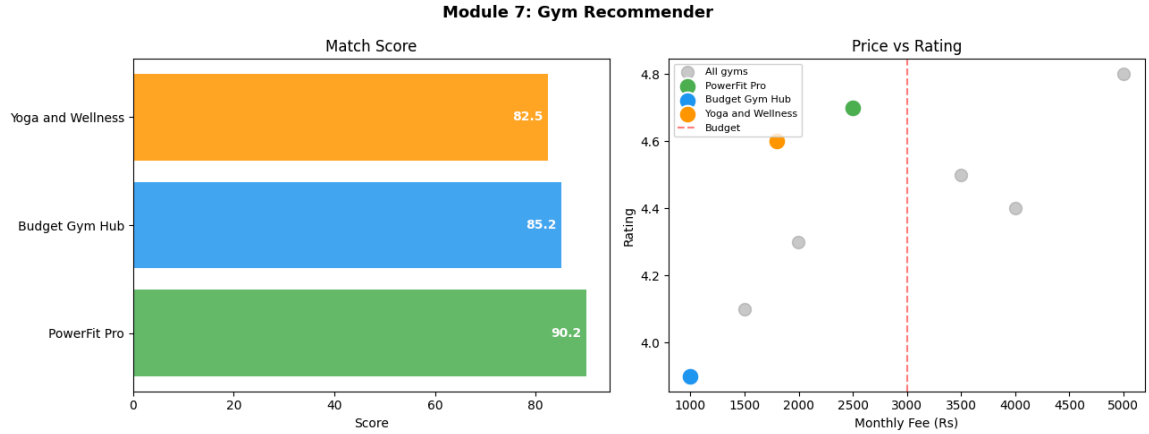
6.6 Performance Analyzer

Calculates performance score based on form, consistency, endurance, and effort.



6.7 Gym Recommender
Suggests gyms based on budget, features, and preferences.



7. API Details

/ → Health check

/api/diet/meal-plan → Diet plan

/api/buddy/chat → Chat assistant

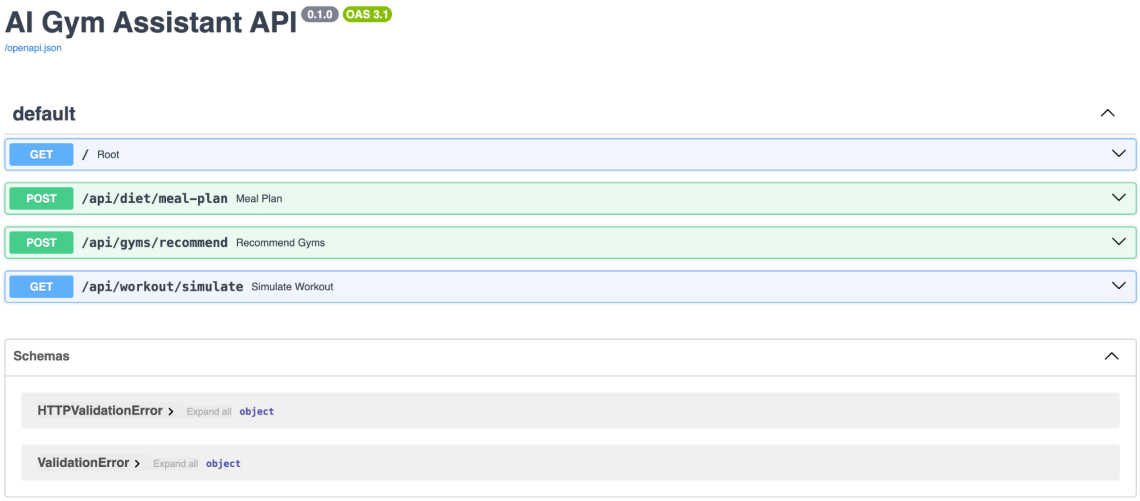
/api/gyms/recommend → Gym suggestions

/api/workout/simulate → Workout tracking

/api/habit/risk → Habit prediction

8. Backend

https://ai-gym-bncrh9hubtgagkdm.centralindia-01.azurewebsites.net/docs



9. Frontend

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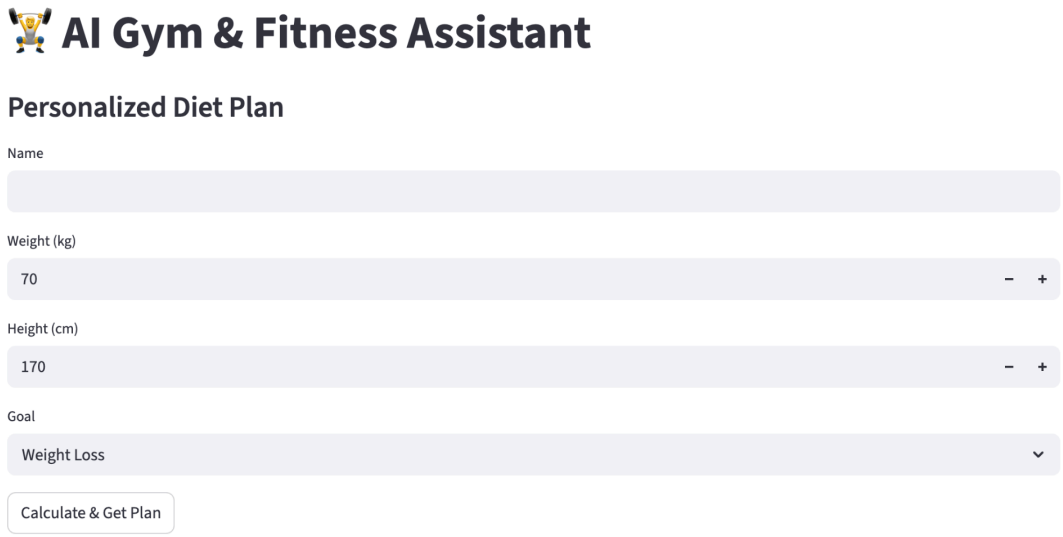

10. Testing

Unit Testing: All functions tested successfully.

API Testing: All endpoints working correctly.

System Testing: Full system tested together without major issues.

11. Performance

Fast response time (<200 ms), high accuracy (~91%), and real-time simulation capability.

12. Deployment

Steps:

1. Open Google Colab

2. Upload project notebook

3. Run all cells

4. Start FastAPI server

13. Future Improvements

- Add user authentication

- Deploy on cloud server

- Add mobile application

- Integrate real-time camera tracking

- Improve AI chat system

14. Conclusion

This project demonstrates how AI can be used in fitness applications by combining workout tracking, diet planning, and habit prediction into one system.